

Project Proposal for Numerical Optimization

Seasonal Model Fitting to Meteorological Data

via Nonlinear Least Squares

Azra Kurić

20113

1 Problem Statement

This project applies the Gauss-Newton method for nonlinear least squares to fit a parametric seasonal model to real monthly temperature records for a Bosnian city (Sarajevo). Monthly mean temperatures follow a clear periodic structure: a dominant annual cycle, a smaller semi-annual harmonic, and a slow linear trend. The goal is to estimate the six parameters of the following model:

$$f(t; \theta) = a_0 + a_1 t + A_1 \sin\left(\frac{2\pi t}{12} + \varphi_1\right) + A_2 \sin\left(\frac{4\pi t}{12} + \varphi_2\right) \quad (1)$$

where t is time in months, a_0 and a_1 define the linear trend, A_1 , A_2 are the sinusoidal amplitudes, and φ_1 , φ_2 are phase offsets. The parameter vector is $\theta = (a_0, a_1, A_1, \varphi_1, A_2, \varphi_2)^T \in \mathbb{R}^6$. Because the model is nonlinear in the phase parameters, linear least squares is insufficient and an iterative approach is required.

Real monthly temperature data for Sarajevo will be sourced from the European Climate Assessment & Dataset (ECA&D) or NOAA's Global Surface Summary of Month (GSOM).

2 Optimization Model

Given N monthly observations $y_1, \dots, y_N$ at times $t_1, \dots, t_N$, the fitting problem is formulated as a nonlinear least-squares minimization:

$$\min_{\theta} F(\theta) = \sum_{i=1}^N r_i(\theta)^2, \quad r_i(\theta) = y_i - f(t_i; \theta) \quad (2)$$

where $r(\theta) \in \mathbb{R}^N$ is the residual vector. Due to the sinusoidal nonlinearity in φ_1 and φ_2 , the objective $F(\theta)$ is non-convex, so the initial parameter estimates are chosen to reflect known physical properties of the Sarajevo climate (e.g. amplitude initialized from the observed temperature range, phase from the known position of summer peak).

3 Numerical Method

The Gauss-Newton method is used, as mentioned in the projects list. At each iteration k , the residual vector is linearized around the current estimate using the Jacobian $J(\theta^k) \in \mathbb{R}^{N \times 6}$, and the normal equations are solved for the update step d^k :

$$J(\theta^k)^T J(\theta^k) d^k = -J(\theta^k)^T r(\theta^k) \quad (3)$$

$$\theta^{k+1} = \theta^k + \alpha_k d^k \quad (4)$$

The step length α_k is determined by the Armijo backtracking line search to ensure sufficient decrease. All six partial derivatives of $f(t_i; \theta)$ with respect to the parameters are computed analytically and assembled into J at each iteration. Iterations continue until the gradient norm and the step norm fall below prescribed tolerances, or a maximum iteration count is reached.

4 Implementation and Visualization

The application will be implemented in C++ using the natID framework. Real temperature data for Sarajevo will be loaded from a CSV file at startup. The graphical interface will allow the user to adjust initial parameter values and convergence settings before running the optimization.

The visualization will include:

- **Main plot:** observed monthly temperature data overlaid with the fitted model curve, updated at each iteration to show how the curve evolves toward the final fit.
- **Residual plot:** a bar chart of $r_i(\theta)$ vs. month index, illustrating the improvement across iterations.
- **Convergence plot:** a log-scale chart of the residual sum of squares $F(\theta^k)$ vs. iteration number.
- **Parameter display:** a table showing the current values of all six components of θ^k updated at each iteration.

An animation mode will step through iterations at a user-controlled rate, allowing the user to observe how the model progressively fits the data.